

Machine Learning - Assignment 3

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Data

Data source: [Google Drive Link](#)

On this assignment, you will work on three datasets. They are:

- **Dry Bean Dataset:** A dataset about classifying the type of beans according to various geometric and structural features. This dataset has been filtered to only include some labels and fewer observations. Do not download the dataset from Kaggle, use the one we provided from the Google Drive link above.
- **Fashion-MNIST:** A dataset consisting of clothing items, represented as 28X28 grayscale pixel values. We have filtered the dataset to only include five specific clothing classes (e.g., T-shirt/top, Pullover, Dress, Coat, Shirt).
- **Synthetic Data (Bonus):** A high-dimensional data synthesized for the bonus section.

IMPORTANT: The data is not currently split into train/test/validation sets. You need to split the data into three sets with a 80/10/10% split (train, test and validation respectively), and train your model with the train set only. Keep in mind the risk of data leakage in each section of the assignment. Differentiate between the train/validation/test sets correctly in the different sections of the assignment. Hyperparameter tuning should only be done on the validation set, and once you find the best parameters, evaluate your model on the test set. This is true for all three sets.

There is no need to implement the models/algorithms you use in this assignment. You may use scikit-learn or any other library, with explanations and annotations.

General experimental design and data leakage

- All preprocessing steps that learn parameters from the data – including imputation, scaling, encoding, PCA, feature selection, and feature dropping – must be fitted only to the training set. Validation and test sets may be transformed only using objects fitted to the training set.
- Hyperparameter tuning must be based on the validation set only. The test set should be used only for final reporting and error analyses after model selection is complete.
- Using scikit-learn Pipeline or ColumnTransformer is strongly recommended whenever preprocessing and modeling are combined.
- For each main modeling comparison, write a brief hypothesis before execution, then compare the hypothesis to the observed results.

- For each supervised-learning task, include a short error analysis of misclassified test examples and discuss what may explain the errors.

Requirements

Part 1 - Dry Bean Dataset (50 pts)

In this part, you would explore few interesting questions about the Dry Bean Dataset:

- Which geometric or shape-related features appear to separate bean classes?
- Do engineered features improve classification or mainly repeat information already present in the original variables?
- Does PCA preserve predictive performance while reducing redundancy?
- Do unsupervised clusters align with the known labels, and where do they fail?

Section A - Data Exploration & Visualization (5 pts)

Explore the data using tables, visualizations, and other relevant methods.

- Plots should have an informative main title, axis labels and a legend (if needed).
- For each plot or table, provide a short description of key observations, including only content which would be meaningful/informative.
- The visualizations should be detailed and cover all relevant aspects of the data.
- The visualizations should highlight any interesting patterns or trends that can be observed in the data, as well as key statistics such as the mean, mode etc. of each feature.
- Include **exactly 5** visualizations using at least **3 different** plot types. Each visualization should include a short explanation of what it shows and how it helps you understand the data or supports later analysis. Avoid adding visualizations that don't add value.

The goal of this section is to get insights on the data which may or may not be relevant for the following sections.

Section B - Data Pre-processing (10 pts)

- Before starting this section, please read Sections C and D to understand the overall goal and guide better preprocessing decisions.
- Apply different methods of pre-processing to the data in order to prepare it for the models you wish to apply in the next sections.

- Perform feature engineering on the data similarly to Assignment 2, generate at least 3 new features, and explain the features you generated.
- Perform other pre-processing steps as we discussed in class (Imputation, transformation etc.) and explain your choices.

Section C - Classification and Clustering (20 pts)

Classification

Use at least three different machine learning models we have studied in class to predict the label of each observation.

- The implementation must include parameter tuning.
- Report a suitable measure to evaluate the performance of each model.
- Present the models' results in a plot, compare the results of the different models, and discuss them.
- Before training, briefly state which classifier you expect to perform best and why. After evaluation, include an error analysis: inspect 3-5 misclassified test beans and compare their values to class-level summaries.

Clustering

Use two different clustering algorithms and attempt to cluster the data **before dimensionality reduction**.

- Clustering is unsupervised learning task. However, you required to not use the labels while fitting the clustering algorithms. After clustering, compare the clusters with the true labels (post hoc).
- Evaluate the clusters using one partitioning evaluation metric (Silhouettes or WCSS) and one simple external evaluation such as a cluster vs label table, majority-label mapping, or purity score.
- To visualize clusters before dimensionality reduction, choose at least two informative feature pairs based on the EDA and plot the observations colored by cluster. Also show or discuss how these clusters correspond to the true labels.

Section D - PCA (15 pts)

Perform Principal Component Analysis on the data.

- Before running PCA, state whether you expect PCA to help or hurt each model type, especially distance-based models such as KNN. After PCA, compare scores, the number of retained components, runtime (when relevant), and error patterns.
- Retain the number of Principal Components (PCs) that explain >80% of the variance (you can modify this value as you wish, but explain why if you choose to do so).

- Perform the same tasks as Section C with the same models/algorithms.
- Compare the results of the lower-dimension data to the results of Section C.
- Discuss differences in performance and clustering to determine why it was better or worse.

Part 2 - Fashion-MNIST (50 pts)

In this part, you would explore few interesting questions about the Fashion-MNIST:

- **Which clothing classes appear visually similar and are therefore likely to be confused ?**
- **Does PCA remove noise and redundancy, or does it remove pixel-level information useful for classification?**
- **Does dropping low-importance pixels improve runtime, accuracy, or both?**
- **Do the most and least important pixels make sense when mapped back to their 228x28 image locations?**

Section A - Visualization (5 pts)

- Visualize several of the observations, so that the clothing item is clearly visible. Visualize at least two of each clothing class.
- Plot a heatmap of the data, showing the average value of each pixel (feature) to see which pixels are the most "popular".
 - Use `.reshape(28, 28)` on the averaged data so the heatmap would show as a 28 X 28 "image".
 - Explain why this visualization is useful for this dataset.

Section B - Classification (15 pts)

Use two machine learning models we have studied in class to train and classify the data (the labels are: 0 - T-shirt/top, 2 - Pullover, 4 - Coat, 6 - Shirt, 8 - Bag).

- The implementation must include parameter tuning.
- Report a suitable measure to evaluate the performance of each model.
- Present the models' results in a plot, compare the results of the different models, and discuss them.
- Log the training time and predict time for each model for comparison later.
- Before training, briefly hypothesize which model will perform better and why. After evaluation, inspect several misclassified test images and explain the likely visual ambiguity. Also compare feature or pixel-intensity distributions for correctly and incorrectly classified examples when useful.

Section C - PCA and Feature Importance/Selection (25 pts)

- Perform PCA in the same manner as Part 1 Section D, logging the training and predict time (not the time it takes to perform PCA).
- After running the models on the reduced-dimension data, compute PCA based feature importance from the retained PCs. Use the suggested formula below, or another reasonable definition (for example, absolute loadings) if clearly explained and used consistently.
 - Suggested formula: $importance_j = \sum_k explained_variance_ratio_k * loading_{(k,j)}^2$. Squared loadings remove sign, since positive and negative loadings can both indicate strong contribution.
- Normalize the feature importance values to sum to 1 before using drop_percent: $normalized_importance_j = \frac{importance_j}{\sum_i importance_i}$.
- For Fashion-MNIST, report feature_index, row, col, and importance for the 5 most important and 5 least important pixels. Visualize their locations on a 28x28 grid.
- Choose a value for drop_percent as a cumulative normalized-importance threshold (not as the percentage of pixels to drop).
- For example, if drop_percent = 0.1, sort the pixels from least important to most important and drop the lowest-importance pixels whose cumulative normalized importance is ≤ 0.1 . The number of dropped pixels depends on the distribution of importance values.
- Report on what features those are, the total number you will drop, and whether these features make sense to drop.
- Drop those features from the original data, then perform sections B and C again (run models on the data after dropping features before and after dimensionality reduction). There is no need to calculate feature importance and selection again, resulting in 4 runs total for each model across sections B and C.
- Compare the results (evaluation metrics and runtime) across all 4 runs and discuss why some were better/worse and if they matched your expectations.
- Before dropping pixels, state whether you expect feature dropping to improve runtime, accuracy, or both. After the four runs, compare the hypothesis to the observed results.

Section D - t-SNE (5 pts)

- Visualize the data using t-SNE in 2D or 3D, coloring each observation of each label in a different color.
- Determine if clusters are apparent in the visualization, and try different hyper-parameters if not.
- If t-SNE is too slow on the full dataset, you may use a stratified sample and/or an initial PCA compression for visualization, but you must report what you did.
- Discuss why trying to visualize clusters without dimensionality reduction, like in Part 1 Section C, would be difficult.

Part 3 – Bonus: LLM/Agent Use Appendix (25 pts)

In this bonus section, you will document and critically evaluate your use of LLMs, coding assistants, or autonomous agents during this assignment.

- A. Include a table with at least three substantial AI/Agent interactions. If you used AI less than three times. For each interaction, report:

Task	Tool used	Prompt/ instruction summary	Output accepted ?	What was changed?	How did you verify it?
Example: Debug KNN pipeline	ChatGPT in browser and Codex	Asked why validation accuracy was unstable	Partially accepted	Rewrote the scaling step inside a Pipeline object	Checked train/validation split, reran with fixed random seed, compared to baseline.

The verification column is the most important part. You should explain how you checked that the AI output was correct.

B. One AI/Agent Failure or Limitation

Describe at least one case where an AI tool gave an incorrect, incomplete, misleading, or overly generic answer. Include:

1. What the AI suggested.
2. Why the suggestion was problematic.
3. How you detected the issue.
4. What you did instead.

C. Give the final solution to AI, ask it to check each part and section, and give you critiques for the full solution. Then report in a table or in a clear markdown cell:

1. What was the prompt you gave to the AI.
2. Which tool it was.
3. For each section, what it suggested. If you accepted or rejected its suggestion, explain exactly why. If you accepted, describe how you implemented its suggestion.

Guidelines

Please read the following section carefully before submitting the assignment.

Coding Guidelines

- Use familiar packages with explicit explanations.
- If you have installed any libraries beyond those presented in the exercises, please specify this in the report.
- The code should run without warnings or errors.
- Good documentation is critical.
- Indicate the exercise sections in the code as well.
- Use meaningful variable names.
- Do not use reserved words.
- Use constants where possible.

Submission Guidelines

- The assignment should be submitted in pairs (only one submission).
- You are required to submit two files including all the sections: one in .ipynb format and one in .html format.
- Both files should also include the program's outputs.
- The files' names should be of the form: ML_HW3_ID1_ID2.
- Assignments submitted late will receive a penalty of 3 points for each day, up to one week, and later submissions will not be accepted.

Grading

- You can get more than 100 points for the exercise.
- The exercise will be graded according to correctness, clarity, leakage-safe implementation, quality of explanations, error analyses, efficiency, elegance of implementation, and responsible use of external tools/LLMs.

Self-learning

- As we mentioned at the beginning of the course, self-learning is an important part of the course.
- Treat all sources of information carefully and critically.
- Usage of LLMs is allowed, but reference it where it was used.
- You can and should consult with other students in the course, but each pair must write their own work.
- It is reasonable to assume that not all results and algorithms will be identical.

Questions and Reception hours

- Please post your questions on the exercise forum in Moodle, after you have read the previous posts.
- Professional questions sent by email will not be answered.
- If you want to schedule a reception hour with one of the instructors, please send your questions by email in advance.
- In any other case (personal questions, request for an extension with a justified reason, etc.) please email the instructor.

Good luck